

Лабораторная работа № 14. Статическая маршрутизация в Интернете. Настройка

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Информация

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Цель работы

Настроить взаимодействие через сеть провайдера посредством статической маршрутизации локальной сети организации с сетью основного здания, расположенного в 42-м квартале в Москве, и сетью филиала, расположенного в г. Сочи.

Задание

1. Настроить связь между территориями.
2. Настроить оборудование, расположенное в квартале 42 в Москве.
3. Настроить оборудование, расположенное в филиале в г. Сочи.
4. Настроить статическую маршрутизацию между территориями.

Задание

5. Настроить статическую маршрутизацию на территории квартала 42 в г. Москве.
6. Настроить NAT на маршрутизаторе msk-donskaya-gw-1.
7. При выполнении работы необходимо учитывать соглашение об именовании.

Выполнение лабораторной работы

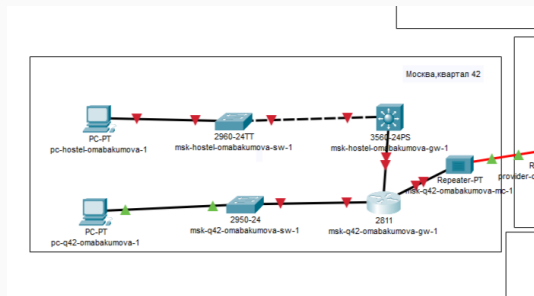


Рис. 1: 42 квартал, Москва

Выполнение лабораторной работы

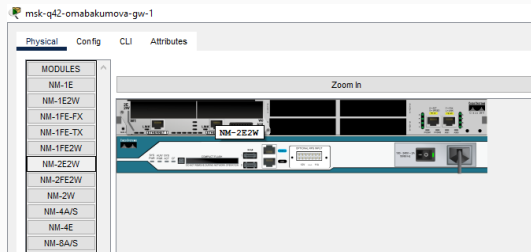


Рис. 2: Ошибочный модуль

Выполнение лабораторной работы

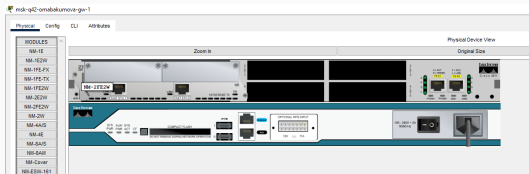


Рис. 3: Правильный модуль

Настройка линка между площадками

```
interface GigabitEthernet0/2
!
interface Vlan1
no ip address
shutdown
!
interface Vlan4
no ip address
!
!
!
!
line con 0
password 7 0822455D0A16
login
!
line vty 0 4
password 7 0822455D0A16
login
line vty 5 15
login
!
!
!
!
end
--More--
```

Рис. 4: Информация о provider-omabakumova-sw-1

Настройка линка между площадками

```
provider-omabakumova-sw-1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
provider-omabakumova-sw-1(config)#interface f0/3
provider-omabakumova-sw-1(config-if)#switchport mode trunk

provider-omabakumova-sw-1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up

provider-omabakumova-sw-1(config-if)#exit
provider-omabakumova-sw-1(config)#interface f0/4
provider-omabakumova-sw-1(config-if)#switchport mode trunk
provider-omabakumova-sw-1(config-if)#exit
provider-omabakumova-sw-1(config)#vlan 5
provider-omabakumova-sw-1(config-vlan)#name q42
provider-omabakumova-sw-1(config-vlan)#exit
provider-omabakumova-sw-1(config)#interface vlan5
provider-omabakumova-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan5, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan5, changed state to up

provider-omabakumova-sw-1(config-if)#no shutdown
provider-omabakumova-sw-1(config-if)#exit
provider-omabakumova-sw-1(config)#vlan 6
provider-omabakumova-sw-1(config-vlan)#name sochi
provider-omabakumova-sw-1(config-vlan)#exit
provider-omabakumova-sw-1(config)#interface vlan6
provider-omabakumova-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan6, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan6, changed state to up

provider-omabakumova-sw-1(config-if)#no shutdown
provider-omabakumova-sw-1(config-if)#exit
provider-omabakumova-sw-1(config)#^Z
provider-omabakumova-sw-1#
%SYS-5-CONFIG_I: Configured from console by console

provider-omabakumova-sw-1#wr mem
Building configuration...
[OK]
provider-omabakumova-sw-1#
```

Рис. 5: Настройка интерфейсов коммутатора provider-omabakumova-sw-1

Настройка линка между площадками

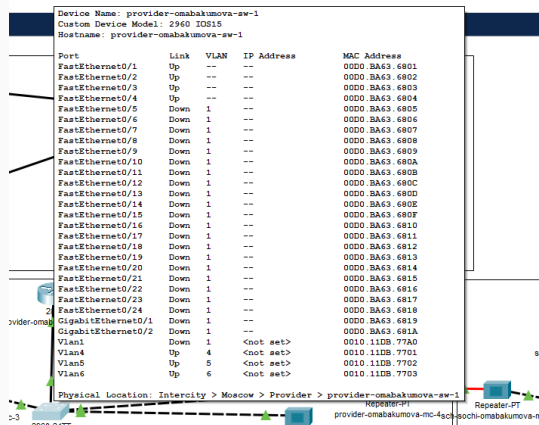


Рис. 6: Успешна настройка provider-omabakumova-sw-1

Настройка линка между площадками

```
msk-donskaya-omabakumova-gw-1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-omabakumova-gw-1(config)#interface f0/1.5
msk-donskaya-omabakumova-gw-1(config-subif)#
%LINK-S-CHANGED: Interface FastEthernet0/1.5, changed state to up

%LINEPROTO-S-UPDOWN: Line protocol on Interface FastEthernet0/1.5, changed state to up

msk-donskaya-omabakumova-gw-1(config-subif)#encapsulation dot1Q 5
msk-donskaya-omabakumova-gw-1(config-subif)#ip address 10.128.255.1 255.255.255.252
msk-donskaya-omabakumova-gw-1(config-subif)#description q42
msk-donskaya-omabakumova-gw-1(config-subif)#exit
msk-donskaya-omabakumova-gw-1(config)#interface f0/1.6
msk-donskaya-omabakumova-gw-1(config-subif)#
%LINK-S-CHANGED: Interface FastEthernet0/1.6, changed state to up

%LINEPROTO-S-UPDOWN: Line protocol on Interface FastEthernet0/1.6, changed state to up

msk-donskaya-omabakumova-gw-1(config-subif)#encapsulation dot1Q 6
msk-donskaya-omabakumova-gw-1(config-subif)#ip address 10.128.255.5 255.255.255.252
msk-donskaya-omabakumova-gw-1(config-subif)#description sochi
msk-donskaya-omabakumova-gw-1(config-subif)#exit
msk-donskaya-omabakumova-gw-1(config)#exit
msk-donskaya-omabakumova-gw-1#
%SYS-S-CONFIG_I: Configured from console by console

msk-donskaya-omabakumova-gw-1#wr mem
Building configuration...
[OK]
msk-donskaya-omabakumova-gw-1#
```

Рис. 7: Настройка интерфейсов маршрутизатора msk-donskaya-omabakumova-gw-1

Настройка линка между площадками

```
:
interface FastEthernet0/1.5
  description q42
  encapsulation dot1Q 5
  ip address 10.128.255.1 255.255.255.252
!
interface FastEthernet0/1.6
  description sochi
  encapsulation dot1Q 6
  ip address 10.128.255.5 255.255.255.252
!
```

Рис. 8: Предварительные результаты настройки msk-donskaya-omabakumova-gw-1

Настройка линка между площадками

```
msk-q42-omabakumova-gw-1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
msk-q42-omabakumova-gw-1(config)#interface f0/1
msk-q42-omabakumova-gw-1(config-if)#no shutdown

msk-q42-omabakumova-gw-1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

msk-q42-omabakumova-gw-1(config-if)#exit
msk-q42-omabakumova-gw-1(config)#interface f0/1.5
msk-q42-omabakumova-gw-1(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/1.5, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1.5, changed state to up

msk-q42-omabakumova-gw-1(config-subif)#encapsulation dot1Q 5
msk-q42-omabakumova-gw-1(config-subif)#ip address 10.128.255.2 255.255.255.252
msk-q42-omabakumova-gw-1(config-subif)#description donskaya
msk-q42-omabakumova-gw-1(config-subif)#exit
msk-q42-omabakumova-gw-1(config)#exit
msk-q42-omabakumova-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-q42-omabakumova-gw-1#wr mem
Building configuration...
[OK]
msk-q42-omabakumova-gw-1#
```

Рис. 9: Настройка интерфейсов маршрутизатора msk-q42-omabakumova-gw-1

Настройка линка между площадками

```
msk-q42-omabakumova-gw-1#ping 10.128.255.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.128.255.1, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/1 ms

msk-q42-omabakumova-gw-1#ping 10.128.255.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.128.255.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/3/18 ms

msk-q42-omabakumova-gw-1#
```

Рис. 10: Пинг успешно проходит с msk-q42-omabakumova-gw-1

Настройка линка между площадками

```
sch-sochi-omabakumova-sw-1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
sch-sochi-omabakumova-sw-1(config)#interface f0/23
sch-sochi-omabakumova-sw-1(config-if)#switchport mode trunk
sch-sochi-omabakumova-sw-1(config-if)#exit
sch-sochi-omabakumova-sw-1(config)#interface f0/24
sch-sochi-omabakumova-sw-1(config-if)#switchport mode trunk
sch-sochi-omabakumova-sw-1(config-if)#exit
sch-sochi-omabakumova-sw-1(config)#vlan 6
sch-sochi-omabakumova-sw-1(config-vlan)#name sochi
sch-sochi-omabakumova-sw-1(config-vlan)#exit
sch-sochi-omabakumova-sw-1(config)#interface vlan6
sch-sochi-omabakumova-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan6, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan6, changed state to up

sch-sochi-omabakumova-sw-1(config-if)#no shutdown
sch-sochi-omabakumova-sw-1(config-if)#exit
sch-sochi-omabakumova-sw-1(config)#
```

Рис. 11: Настройка интерфейсов коммутатора sch-sochi-omabakumova-sw-1

Настройка линка между площадками

```
interface FastEthernet0/23
  switchport mode trunk
!
interface FastEthernet0/24
  switchport mode trunk
!
interface Vlan1
  no ip address
  shutdown
!
interface Vlan6
  no ip address
!
.
```

Рис. 12: Предварительные результаты настройки sch-sochi-omabakumova-sw-1

Настройка линка между площадками

```
sch-sochi-omabakumova-gw-1> enable
sch-sochi-omabakumova-gw-1# configure terminal
sch-sochi-omabakumova-gw-1(config)# interface f0/0
sch-sochi-omabakumova-gw-1(config-if)#no shutdown
sch-sochi-omabakumova-gw-1(config-if)#exit
sch-sochi-omabakumova-gw-1(config)# interface f0 /0.6
sch-sochi-omabakumova-gw-1(config-subif)# encapsulation dot1Q 6
sch-sochi-omabakumova-gw-1(config-subif)#ip address 10.128.255.6 255.255.255.6
sch-sochi-omabakumova-gw-1(config-subif)# description donskaya
sch-sochi-omabakumova-gw-1(config-subif)#exit
sch-sochi-omabakumova-gw-1(config)#exit
```

Настройка линка между площадками

```
interface FastEthernet0/0
  no ip address
  duplex auto
  speed auto
!
interface FastEthernet0/0.6
  description donskaya
  encapsulation dot1Q 6
  ip address 10.128.255.6 255.255.255.252
```

Рис. 13: Предварительные результаты настройки sch-sochi-omabakumova-gw-1

Настройка линка между площадками

```
sch-sochi-omabakumova-gw-1#ping 10.128.255.5
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.128.255.5, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/1 ms

sch-sochi-omabakumova-gw-1#ping 10.128.255.5
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.128.255.5, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/3/18 ms

sch-sochi-omabakumova-gw-1#
```

Рис. 14: Пинг успешно проходит с sch-sochi-omabakumova-gw-1

Настройка площадки 42-го квартала

```
msk-q42-omabakumova-gw-1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
msk-q42-omabakumova-gw-1(config)#interface f0/0
msk-q42-omabakumova-gw-1(config-if)#no shutdown

msk-q42-omabakumova-gw-1(config-if)#
%LINEPROTO-5-UPDOWN: Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

msk-q42-omabakumova-gw-1(config-if)#exit
msk-q42-omabakumova-gw-1(config)#interface f0/0.201
msk-q42-omabakumova-gw-1(config-subif)#
%LINK-6-CHANGED: Interface FastEthernet0/0.201, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.201, changed state to up

msk-q42-omabakumova-gw-1(config-subif)#encapsulation dot1Q 201
msk-q42-omabakumova-gw-1(config-subif)#ip address 10.129.0.1 255.255.255.0
msk-q42-omabakumova-gw-1(config-subif)#description q42main
msk-q42-omabakumova-gw-1(config-subif)#
msk-q42-omabakumova-gw-1(config-subif)#exit
msk-q42-omabakumova-gw-1(config)#interface f1/0
msk-q42-omabakumova-gw-1(config-if)#no shutdown
msk-q42-omabakumova-gw-1(config-if)#exit
msk-q42-omabakumova-gw-1(config)#interface f1/0.202
msk-q42-omabakumova-gw-1(config-subif)#
%LINK-6-CHANGED: Interface FastEthernet1/0.202, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0.202, changed state to up

msk-q42-omabakumova-gw-1(config-subif)#encapsulation dot1Q 202
msk-q42-omabakumova-gw-1(config-subif)#ip address 10.129.1.1 255.255.255.0
msk-q42-omabakumova-gw-1(config-subif)#description q42management

% Invalid input detected at '' marker.

msk-q42-omabakumova-gw-1(config-subif)#description q42-management
% Invalid input detected at '' marker.

msk-q42-omabakumova-gw-1(config-subif)#description q42-management
msk-q42-omabakumova-gw-1(config-subif)#exit
msk-q42-omabakumova-gw-1(config)#^Z
msk-q42-omabakumova-gw-1#
MSR0-6-CONFIG_1: Configured from console by console

msk-q42-omabakumova-gw-1#r men
Building configuration...
[OK]
msk-q42-omabakumova-gw-1#
```

Рис. 15: Настройка интерфейсов маршрутизатора msk-q42-omabakumova-gw-1 для настройки площадки 42 квартала

Настройка площадки 42-го квартала

```
msk-q42-omabakumova-sw-1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
msk-q42-omabakumova-sw-1(config)#interface f0/24
msk-q42-omabakumova-sw-1(config-if)#switchport mode trunk
msk-q42-omabakumova-sw-1(config-if)#exit
msk-q42-omabakumova-sw-1(config)#interface f0/1
msk-q42-omabakumova-sw-1(config-if)#switchport mode access
msk-q42-omabakumova-sw-1(config-if)#switchport access vlan 201
% Access VLAN does not exist. Creating vlan 201
msk-q42-omabakumova-sw-1(config-if)#exit
msk-q42-omabakumova-sw-1(config)#vlan 201
msk-q42-omabakumova-sw-1(config-vlan)#name q42-main
msk-q42-omabakumova-sw-1(config-vlan)#exit
msk-q42-omabakumova-sw-1(config)#interface vlan201
msk-q42-omabakumova-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan201, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan201, changed state to up

msk-q42-omabakumova-sw-1(config-if)#no shutdown
msk-q42-omabakumova-sw-1(config-if)#exit
msk-q42-omabakumova-sw-1(config)#^Z
msk-q42-omabakumova-sw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-q42-omabakumova-sw-1#wr mem
Building configuration...
[OK]
msk-q42-omabakumova-sw-1#
```

Рис. 16: Настройка интерфейсов коммутатора msk-q42-omabakumova-sw-1

Настройка площадки 42-го квартала

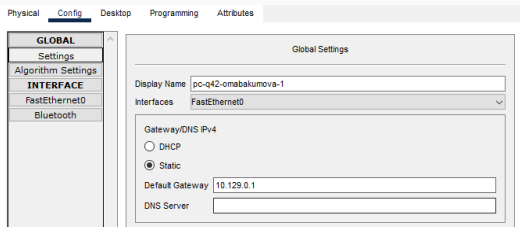


Рис. 17: Задание шлюза у pc-q42-omabakumova-1

Настройка площадки 42-го квартала

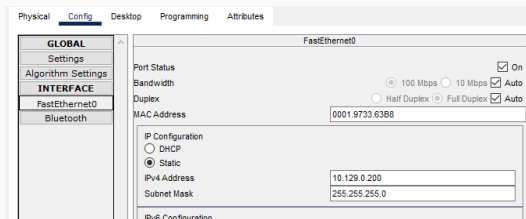
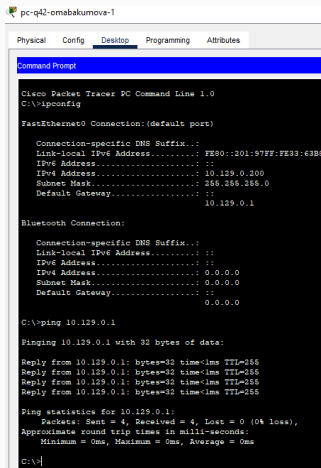


Рис. 18: Задание статического адреса у pc-q42-omabakumova-1

Настройка площадки 42-го квартала



```
pc-q42-omabakumova-1
Physical  Config  Desktop  Programming  Attributes
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig

FastEthernet0 Connection: (default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::201:97FF:FE33:63B8
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 10.129.0.200
    Subnet Mask . . . . .: 256.256.256.0
    Default Gateway . . . . .: ::
                                10.129.0.1

Bluetooth Connection:

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: ::
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 0.0.0.0
    Subnet Mask . . . . .: 0.0.0.0
    Default Gateway . . . . .: ::
                                0.0.0.0

C:\>ping 10.129.0.1

Pinging 10.129.0.1 with 32 bytes of data:

Reply from 10.129.0.1: bytes=32 time<1ms TTL=255
Reply from 10.129.0.1: bytes=32 time<1ms TTL=255
Reply from 10.129.0.1: bytes=32 time<1ms TTL=255
Reply from 10.129.0.1: bytes=32 time<1ms TTL=255

Ping statistics for 10.129.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Рис. 19: Пингование в пределах 42 квартала

Настройка площадки 42-го квартала

```
C:\>ping 10.128.0.1

Pinging 10.128.0.1 with 32 bytes of data:

Reply from 10.129.0.1: Destination host unreachable.
Reply from 10.129.0.1: Destination host unreachable.
Reply from 10.129.0.1: Destination host unreachable.
Reply from 10.129.0.1: Destination host unreachable.

Ping statistics for 10.128.0.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

Рис. 20: Попытка пустить пинг за пределы области

Настройка площадки 42-го квартала

```
msk-hostel-omabakumova-gw-1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
msk-hostel-omabakumova-gw-1(config)#interface g0/1
msk-hostel-omabakumova-gw-1(config-if)#switchport trunk encapsulation dot1q
msk-hostel-omabakumova-gw-1(config-if)#switchport mode trunk

msk-hostel-omabakumova-gw-1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

msk-hostel-omabakumova-gw-1(config-if)#exit
msk-hostel-omabakumova-gw-1(config)#interface f0/1
msk-hostel-omabakumova-gw-1(config-if)#switchport trunk encapsulation dot1q
msk-hostel-omabakumova-gw-1(config-if)#switchport mode trunk

msk-hostel-omabakumova-gw-1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

msk-hostel-omabakumova-gw-1(config-if)#exit
msk-hostel-omabakumova-gw-1(config)#vlan 202
msk-hostel-omabakumova-gw-1(config-vlan)#name q02management
msk-hostel-omabakumova-gw-1(config-vlan)#name q02management
msk-hostel-omabakumova-gw-1(config-vlan)#name q02-management
msk-hostel-omabakumova-gw-1(config-vlan)#exit
msk-hostel-omabakumova-gw-1(config)#interface vlan202
msk-hostel-omabakumova-gw-1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan202, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan202, changed state to up

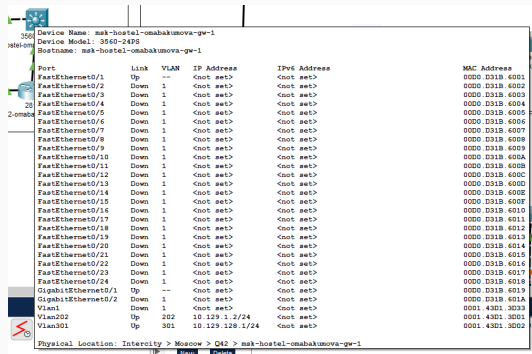
msk-hostel-omabakumova-gw-1(config-if)#no shutdown
msk-hostel-omabakumova-gw-1(config-if)#ip address 10.129.1.2 255.255.255.0
msk-hostel-omabakumova-gw-1(config-if)#exit
msk-hostel-omabakumova-gw-1(config)#vlan 301
msk-hostel-omabakumova-gw-1(config-vlan)#name hostel-main
msk-hostel-omabakumova-gw-1(config-vlan)#exit
msk-hostel-omabakumova-gw-1(config)#interface vlan301
msk-hostel-omabakumova-gw-1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan301, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan301, changed state to up

msk-hostel-omabakumova-gw-1(config-if)#no shutdown
msk-hostel-omabakumova-gw-1(config-if)#ip address 10.129.120.1 255.255.255.0
msk-hostel-omabakumova-gw-1(config-if)#exit
msk-hostel-omabakumova-gw-1(config)#^Z
msk-hostel-omabakumova-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-hostel-omabakumova-gw-1#scr mm
Building configuration...
100%
msk-hostel-omabakumova-gw-1#
```

Рис. 21: Настройка интерфейсов маршрутизирующего коммутатора msk-hostel-omabaumova-gw-1

Настройка площадки 42-го квартала



Port	Link	VLAN	IP Address	IPv6 Address	MAC Address
FastEthernet0/1	Up	--	<not set>	<not set>	0000.D31B.6001
FastEthernet0/2	Down	1	<not set>	<not set>	0000.D31B.6002
FastEthernet0/3	Down	1	<not set>	<not set>	0000.D31B.6003
FastEthernet0/4	Down	1	<not set>	<not set>	0000.D31B.6004
FastEthernet0/5	Down	1	<not set>	<not set>	0000.D31B.6005
FastEthernet0/6	Down	1	<not set>	<not set>	0000.D31B.6006
FastEthernet0/7	Down	1	<not set>	<not set>	0000.D31B.6007
FastEthernet0/8	Down	1	<not set>	<not set>	0000.D31B.6008
FastEthernet0/9	Down	1	<not set>	<not set>	0000.D31B.6009
FastEthernet0/10	Down	1	<not set>	<not set>	0000.D31B.600A
FastEthernet0/11	Down	1	<not set>	<not set>	0000.D31B.600B
FastEthernet0/12	Down	1	<not set>	<not set>	0000.D31B.600C
FastEthernet0/13	Down	1	<not set>	<not set>	0000.D31B.600D
FastEthernet0/14	Down	1	<not set>	<not set>	0000.D31B.600E
FastEthernet0/15	Down	1	<not set>	<not set>	0000.D31B.600F
FastEthernet0/16	Down	1	<not set>	<not set>	0000.D31B.6010
FastEthernet0/17	Down	1	<not set>	<not set>	0000.D31B.6011
FastEthernet0/18	Down	1	<not set>	<not set>	0000.D31B.6012
FastEthernet0/19	Down	1	<not set>	<not set>	0000.D31B.6013
FastEthernet0/20	Down	1	<not set>	<not set>	0000.D31B.6014
FastEthernet0/21	Down	1	<not set>	<not set>	0000.D31B.6015
FastEthernet0/22	Down	1	<not set>	<not set>	0000.D31B.6016
FastEthernet0/23	Down	1	<not set>	<not set>	0000.D31B.6017
FastEthernet0/24	Down	1	<not set>	<not set>	0000.D31B.6018
GigabitEthernet0/1	Up	--	<not set>	<not set>	0000.D31B.6019
GigabitEthernet0/2	Down	1	<not set>	<not set>	0000.D31B.601A
Vlan1	Down	1	<not set>	<not set>	0001.43D1.3D03
Vlan202	Up	202	10.129.1.2/24	<not set>	0001.43D1.3D01
Vlan301	Up	301	10.129.128.1/24	<not set>	0001.43D1.3D02

Physical Location: Intercity > Moscow > Q42 > msk-hostel-omabakumova-gw-1

Рис. 22: проверка проведенной настройки маршрутизирующего коммутатора msk-hostel-omabaumova-gw-1

Настройка площадки 42-го квартала

```
msh-q42-omabakumova-gw-1>en
Password:
msh-q42-omabakumova-gw-1#ping 10.129.1.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.129.1.2, timeout is 2 seconds:
..!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/4/18 ms

msh-q42-omabakumova-gw-1#ping 10.129.1.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.129.1.2, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms

msh-q42-omabakumova-gw-1#
```

Рис. 23: Пинг успешно проходит к msk-hostel-omabaumova-gw-1

Настройка площадки 42-го квартала

```
msk-hostel-omabakumova-sw-1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
msk-hostel-omabakumova-sw-1(config)#interface g0/1
msk-hostel-omabakumova-sw-1(config-if)#switchport mode trunk
msk-hostel-omabakumova-sw-1(config-if)#exit
msk-hostel-omabakumova-sw-1(config)#interface f0/1
msk-hostel-omabakumova-sw-1(config-if)#switchport mode access
msk-hostel-omabakumova-sw-1(config-if)#switchport access vlan 301
% Access VLAN does not exist. Creating vlan 301
msk-hostel-omabakumova-sw-1(config-if)#exit
msk-hostel-omabakumova-sw-1(config)#vlan 301
msk-hostel-omabakumova-sw-1(config-vlan)#name hostel-main
msk-hostel-omabakumova-sw-1(config-vlan)#exit
msk-hostel-omabakumova-sw-1(config)#interface vlan301
msk-hostel-omabakumova-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan301, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan301, changed state to up

msk-hostel-omabakumova-sw-1(config-if)#no shutdown
msk-hostel-omabakumova-sw-1(config-if)#exit
msk-hostel-omabakumova-sw-1(config)#^Z
msk-hostel-omabakumova-sw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-hostel-omabakumova-sw-1#wr mem
Building configuration...
[OK]
msk-hostel-omabakumova-sw-1#
```

Рис. 24: Настройка интерфейсов коммутатора msk-hostel-omabakumova-sw-1

Настройка площадки 42-го квартала

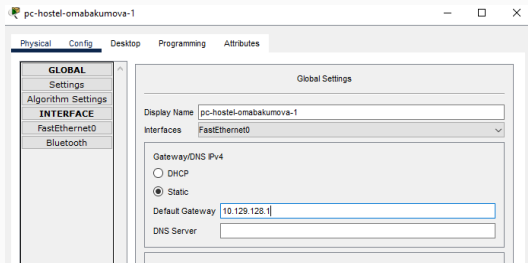


Рис. 25: Задание шлюза у pc-hostel-omabakumova-1

Настройка площадки 42-го квартала

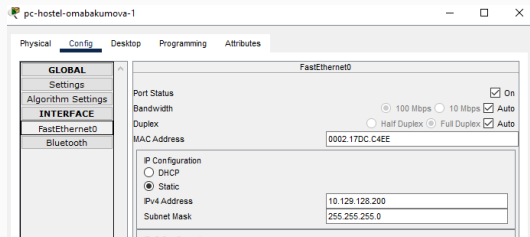
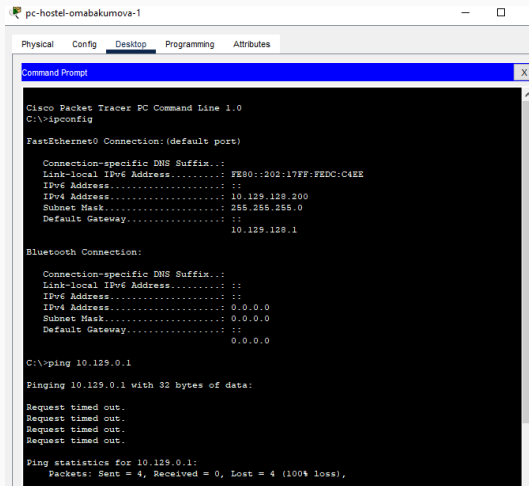


Рис. 26: Задание статического адреса у pc-hostel-omabakumova-1

Настройка площадки 42-го квартала



```
pc-hostel-omabakumova-1
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig

FastEthernet0 Connection: (default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::202:17FF:FEDC:C4EE
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 10.129.128.200
    Subnet Mask . . . . .: 255.255.255.0
    Default Gateway . . . . .: ::
                                10.129.128.1

Bluetooth Connection:

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: ::
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 0.0.0.0
    Subnet Mask . . . . .: 0.0.0.0
    Default Gateway . . . . .: ::
                                0.0.0.0

C:\>ping 10.129.0.1

Pinging 10.129.0.1 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.129.0.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Рис. 27: Попытка пингования с pc-hostel-omabakumova-1

Настройка площадки в Сочи

```
sch-sochi-omabakumova-gw-1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
sch-sochi-omabakumova-gw-1(config)#interface f0/0.401
sch-sochi-omabakumova-gw-1(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/0.401, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.401, changed state to up

sch-sochi-omabakumova-gw-1(config-subif)#encapsulation dot1Q 401
sch-sochi-omabakumova-gw-1(config-subif)#ip address 10.130.0.1 255.255.255.0
sch-sochi-omabakumova-gw-1(config-subif)#description sochi-main
sch-sochi-omabakumova-gw-1(config-subif)#exit
sch-sochi-omabakumova-gw-1(config)#interface f0/0.402
sch-sochi-omabakumova-gw-1(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/0.402, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.402, changed state to up

sch-sochi-omabakumova-gw-1(config-subif)#encapsulation dot1Q 402
sch-sochi-omabakumova-gw-1(config-subif)#ip address 10.130.1.1 255.255.255.0
sch-sochi-omabakumova-gw-1(config-subif)#description sochi-management
sch-sochi-omabakumova-gw-1(config-subif)#exit
sch-sochi-omabakumova-gw-1(config)#^Z
sch-sochi-omabakumova-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

sch-sochi-omabakumova-gw-1#wr mem
Building configuration...
[OK]
sch-sochi-omabakumova-gw-1#
```

Рис. 28: Настройка интерфейсов маршрутизатора sch-sochi-omabakumova-gw-1

Настройка площадки в Сочи

```
sch-sochi-omabakumova-sw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
sch-sochi-omabakumova-sw-1(config)#interface f0/1
sch-sochi-omabakumova-sw-1(config-if)#switchport mode access
sch-sochi-omabakumova-sw-1(config-if)#switchport access vlan 401
% Access VLAN does not exist. Creating vlan 401
sch-sochi-omabakumova-sw-1(config-if)#exit
sch-sochi-omabakumova-sw-1(config)#vlan 401
sch-sochi-omabakumova-sw-1(config-vlan)#name sochi-main
sch-sochi-omabakumova-sw-1(config-vlan)#exit
sch-sochi-omabakumova-sw-1(config)#interface vlan401
sch-sochi-omabakumova-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan401, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan401, changed state to up

sch-sochi-omabakumova-sw-1(config-if)#no shutdown
sch-sochi-omabakumova-sw-1(config-if)#exit
sch-sochi-omabakumova-sw-1(config)#^Z
sch-sochi-omabakumova-sw-1#
%SYS-5-CONFIG_I: Configured from console by console

sch-sochi-omabakumova-sw-1#wr mem
Building configuration...
[OK]
sch-sochi-omabakumova-sw-1#
```

Рис. 29: Настройка интерфейсов коммутатора sch-sochi-omabakumova-sw-1

Настройка площадки в Сочи

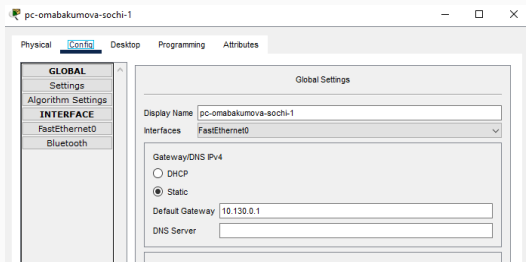


Рис. 30: Задание шлюза у pc-omabakumova-sochi-1

Настройка площадки в Сочи

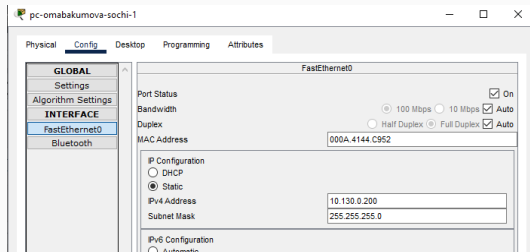


Рис. 31: Задание статического адреса у pc-omabakumova-sochi-1

Настройка маршрутизации между площадками

```
msk-donskaya-omabakumova-gw-1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-omabakumova-gw-1(config)#ip route 10.129.0.0 255.255.0.0 10.128.255.2
msk-donskaya-omabakumova-gw-1(config)#ip route 10.130.0.0 255.255.0.0 10.128.255.6
msk-donskaya-omabakumova-gw-1(config)#^Z
msk-donskaya-omabakumova-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-omabakumova-gw-1#wr mem
Building configuration...
[OK]
msk-donskaya-omabakumova-gw-1#ping 10.129.0.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.129.0.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/1 ms

msk-donskaya-omabakumova-gw-1#
```

Рис. 32: Настройка маршрутизатора msk-donskaya-omabakumova-gw-1

Настройка маршрутизации между площадками

```
msh-donskaya-omabakumova-gw-1#ping 10.129.0.200
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.129.0.200, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/4/19 ms
msh-donskaya-omabakumova-gw-1#
```

Рис. 33: Пинг оконечного устройства 42 квартала

Настройка маршрутизации между площадками

```

admin
Physical      Config      Connectivity  Programming  Attributes

Command Prompt

C:\>ipconfig

C:\>netsh wlan show wlanapi
WlanAPI
Connection-specific BSS Id: {...}
Local-Link Local Address: {...}
IPV4 Address: {...}
IPV6 Address: {...}
Subnet Mask: {...}
Default Gateway: {...}

WlanAPI
Connection-specific BSS Id: {...}
Local-Link Local Address: {...}
IPV4 Address: {...}
IPV6 Address: {...}
Subnet Mask: {...}
Default Gateway: {...}

C:\>ipconfig 19.128.0.1

Flashing 19.128.0.1 with 32 bytes of data:

Reply from 19.128.0.1: bytes=32 time=11ms TTL=64
Request timeout time is 1000 ms.
Status: Success - Success - Success - Success - Success

C:\>ipconfig 19.128.255.1

Flashing 19.128.255.1 with 32 bytes of data:

Reply from 19.128.255.1: bytes=32 time=11ms TTL=64
Reply from 19.128.255.1: bytes=32 time=11ms TTL=64
Reply from 19.128.255.1: bytes=32 time=11ms TTL=64
Reply from 19.128.255.1: bytes=32 time=11ms TTL=64

Reply successful for 19.128.255.1:
Request: Success - S, Success - S, Lost = 0 (100 Success),
Appropriate round trip time is 10 ms, Average = 10 ms

C:\>ipconfig 19.128.255.3

Flashing 19.128.255.3 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Reply successful for 19.128.255.3:
Request: Success - S, Success - S, Lost = 0 (100% Success),

```

Рис. 34: Пингование различных устройств

Настройка маршрутизации между площадками

```
msk-q42-omabakumova-gw-1>en
Password:
msk-q42-omabakumova-gw-1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
msk-q42-omabakumova-gw-1(config)#ip route 0.0.0.0 0.0.0.0 10.128.255.1
msk-q42-omabakumova-gw-1(config)#^Z
msk-q42-omabakumova-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-q42-omabakumova-gw-1#wr mem
Building configuration...
[OK]
msk-q42-omabakumova-gw-1#
```

Рис. 35: Настройка маршрутизатора msk-q42-omabakumova-gw-1

Настройка маршрутизации между площадками

```
C:\>ping 10.128.255.2

Pinging 10.128.255.2 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.128.255.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 10.128.255.2

Pinging 10.128.255.2 with 32 bytes of data:

Reply from 10.128.255.2: bytes=32 time<1ms TTL=254
Reply from 10.128.255.2: bytes=32 time<1ms TTL=254
Reply from 10.128.255.2: bytes=32 time<1ms TTL=254
Reply from 10.128.255.2: bytes=32 time<1ms TTL=254

Ping statistics for 10.128.255.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Рис. 36: Повторная попытка пинования

Настройка маршрутизации между площадками

```
C:\>ping 10.129.0.1

Pinging 10.129.0.1 with 32 bytes of data:

Reply from 10.129.0.1: bytes=32 time<1ms TTL=254
Reply from 10.129.0.1: bytes=32 time=2ms TTL=254
Reply from 10.129.0.1: bytes=32 time<1ms TTL=254
Reply from 10.129.0.1: bytes=32 time<1ms TTL=254

Ping statistics for 10.129.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 2ms, Average = 0ms

C:\>ping 10.129.0.200

Pinging 10.129.0.200 with 32 bytes of data:

Reply from 10.129.0.200: bytes=32 time=1ms TTL=126
Reply from 10.129.0.200: bytes=32 time<1ms TTL=126
Reply from 10.129.0.200: bytes=32 time<1ms TTL=126
Reply from 10.129.0.200: bytes=32 time<1ms TTL=126

Ping statistics for 10.129.0.200:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

Рис. 37: Пингование устройств 42 квартала

Настройка маршрутизации между площадками

```
sch-sochi-omabakumova-gw-1>en
Password:
sch-sochi-omabakumova-gw-1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
sch-sochi-omabakumova-gw-1(config)#ip route 0.0.0.0 0.0.0.0 10.128.255.5
sch-sochi-omabakumova-gw-1(config)#^Z
sch-sochi-omabakumova-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

sch-sochi-omabakumova-gw-1#wr mem
Building configuration...
[OK]
sch-sochi-omabakumova-gw-1#
```

Рис. 38: Настройка маршрутизатора sch-sochi-omabakumova-gw-1

Настройка маршрутизации на 42 квартале

```
msk-q42-omabakumova-gw-1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
msk-q42-omabakumova-gw-1(config)#ip route 10.129.128.0 255.255.128.0 10.129.1.2
msk-q42-omabakumova-gw-1(config)#^Z
msk-q42-omabakumova-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-q42-omabakumova-gw-1#wr mem
Building configuration...
[OK]
msk-q42-omabakumova-gw-1#
```

Рис. 39: Настройка маршрутизации msk-q42-omabakumova-gw-1

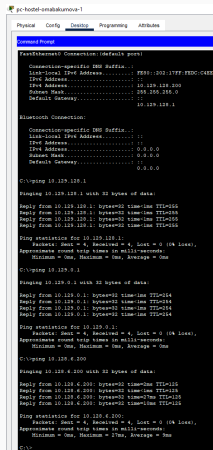
Настройка маршрутизации на 42 квартале

```
msk-hostel-omabakumova-gw-1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
msk-hostel-omabakumova-gw-1(config)#ip routing
msk-hostel-omabakumova-gw-1(config)#ip route 0.0.0.0 0.0.0.0 10.129.1.1
msk-hostel-omabakumova-gw-1(config)#^Z
msk-hostel-omabakumova-gw-1#
%SYS-S-CONFIG_I: Configured from console by console

msk-hostel-omabakumova-gw-1#wr mem
Building configuration...
[OK]
msk-hostel-omabakumova-gw-1#
```

Рис. 40: Настройка маршрутизации msk-hostel-omabakumova-gw-1

Настройка маршрутизации на 42 квартале



```
pc-hostel-smlakunova-1
Physical  Config  Desktop  Programming  Attributes

Command Window
Command Window

FastEthernet0/24 Connection: (default port)
Connection-specific DNS Suffix...:
Link-local IPv6 Address...: FE80::202::17FF:FEDC::C48E
IPv6 Address...:
IPv4 Address...: 10.129.128.200
Subnet Mask...: 255.255.255.0
Default Gateway...:
10.129.128.1

Bluetooth Connection:
Connection-specific DNS Suffix...:
Link-local IPv6 Address...:
IPv6 Address...:
IPv4 Address...: 0.0.0.0
Subnet Mask...: 0.0.0.0
Default Gateway...:
0.0.0.0

C:\>ping 10.129.128.1

Pinging 10.129.128.1 with 32 bytes of data:
Reply from 10.129.128.1: bytes=32 time=1ms TTL=64
Reply from 10.129.128.1: bytes=32 time=1ms TTL=64
Reply from 10.129.128.1: bytes=32 time=1ms TTL=64
Reply from 10.129.128.1: bytes=32 time=1ms TTL=64

Ping statistics for 10.129.128.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milliseconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 10.129.0.1

Pinging 10.129.0.1 with 32 bytes of data:
Reply from 10.129.0.1: bytes=32 time=1ms TTL=64
Reply from 10.129.0.1: bytes=32 time=1ms TTL=64
Reply from 10.129.0.1: bytes=32 time=1ms TTL=64
Reply from 10.129.0.1: bytes=32 time=1ms TTL=64

Ping statistics for 10.129.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milliseconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 10.129.6.200

Pinging 10.129.6.200 with 32 bytes of data:
Reply from 10.129.6.200: bytes=32 time=1ms TTL=64
Reply from 10.129.6.200: bytes=32 time=1ms TTL=64
Reply from 10.129.6.200: bytes=32 time=1ms TTL=64
Reply from 10.129.6.200: bytes=32 time=1ms TTL=64

Ping statistics for 10.129.6.200:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milliseconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

Рис. 41: Успешное пингование устройств

```
msk-donskaya-omabakumova-gw-1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-omabakumova-gw-1(config)#interface f0/1.5
msk-donskaya-omabakumova-gw-1(config-subif)#ip nat inside
msk-donskaya-omabakumova-gw-1(config-subif)#exit
msk-donskaya-omabakumova-gw-1(config)#interface f0/1.6
msk-donskaya-omabakumova-gw-1(config-subif)#ip nat inside
msk-donskaya-omabakumova-gw-1(config-subif)#exit
msk-donskaya-omabakumova-gw-1(config)#ip access-list extended nat-inet
msk-donskaya-omabakumova-gw-1(config-ext-nacl)#remark q42
msk-donskaya-omabakumova-gw-1(config-ext-nacl)#permit ip host 10.129.0.200 any
msk-donskaya-omabakumova-gw-1(config-ext-nacl)#permit ip host 10.129.128.200 any
msk-donskaya-omabakumova-gw-1(config-ext-nacl)#remark sochi
msk-donskaya-omabakumova-gw-1(config-ext-nacl)#permit ip host 10.130.0.200 any
msk-donskaya-omabakumova-gw-1(config-ext-nacl)#exit
msk-donskaya-omabakumova-gw-1(config)#^Z
msk-donskaya-omabakumova-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-omabakumova-gw-1#wr mem
Building configuration...
[OK]
msk-donskaya-omabakumova-gw-1#
```

Рис. 42: Настройка NAT на маршрутизаторе msk-donskaya-omabakumova-gw-1

```
interface FastEthernet0/1.5
  description q42
  encapsulation dot1Q 5
  ip address 10.128.255.1 255.255.255.252
  ip nat inside
!
interface FastEthernet0/1.6
  description sochi
  encapsulation dot1Q 6
  ip address 10.128.255.5 255.255.255.252
  ip nat inside
!
interface Wan1
```

Рис. 43: Результаты предварительной настройки NAT на маршрутизаторе msk-donskaya-omabakumova-gw-1

```
C:\>ping 192.0.2.11

Pinging 192.0.2.11 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Reply from 192.0.2.11: bytes=32 time<1ms TTL=125

Ping statistics for 192.0.2.11:
    Packets: Sent = 4, Received = 1, Lost = 3 (75% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.0.2.11

Pinging 192.0.2.11 with 32 bytes of data:

Reply from 192.0.2.11: bytes=32 time=1ms TTL=125
Reply from 192.0.2.11: bytes=32 time=1ms TTL=125
Reply from 192.0.2.11: bytes=32 time=1ms TTL=125
Reply from 192.0.2.11: bytes=32 time=1ms TTL=125

Ping statistics for 192.0.2.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 1ms, Average = 1ms

C:\>
```

Рис. 44: Пингование www.yandex.ru

```
C:\>ping 192.0.2.14

Pinging 192.0.2.14 with 32 bytes of data:

Request timed out.
Reply from 192.0.2.14: bytes=32 time<1ms TTL=125
Reply from 192.0.2.14: bytes=32 time<1ms TTL=125
Reply from 192.0.2.14: bytes=32 time<1ms TTL=125

Ping statistics for 192.0.2.14:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 192.0.2.14

Pinging 192.0.2.14 with 32 bytes of data:

Reply from 192.0.2.14: bytes=32 time<1ms TTL=125
Reply from 192.0.2.14: bytes=32 time<1ms TTL=125
Reply from 192.0.2.14: bytes=32 time<1ms TTL=125
Reply from 192.0.2.14: bytes=32 time<1ms TTL=125

Ping statistics for 192.0.2.14:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

Рис. 45: Пингование www.rudn.ru

Выводы

В процессе выполнения лабораторной работы я настроила взаимодействие через сеть провайдера посредством статической маршрутизации локальной сети организации с сетью основного здания, расположенного в 42-м квартале в Москве, и сетью филиала, расположенного в г. Сочи.